

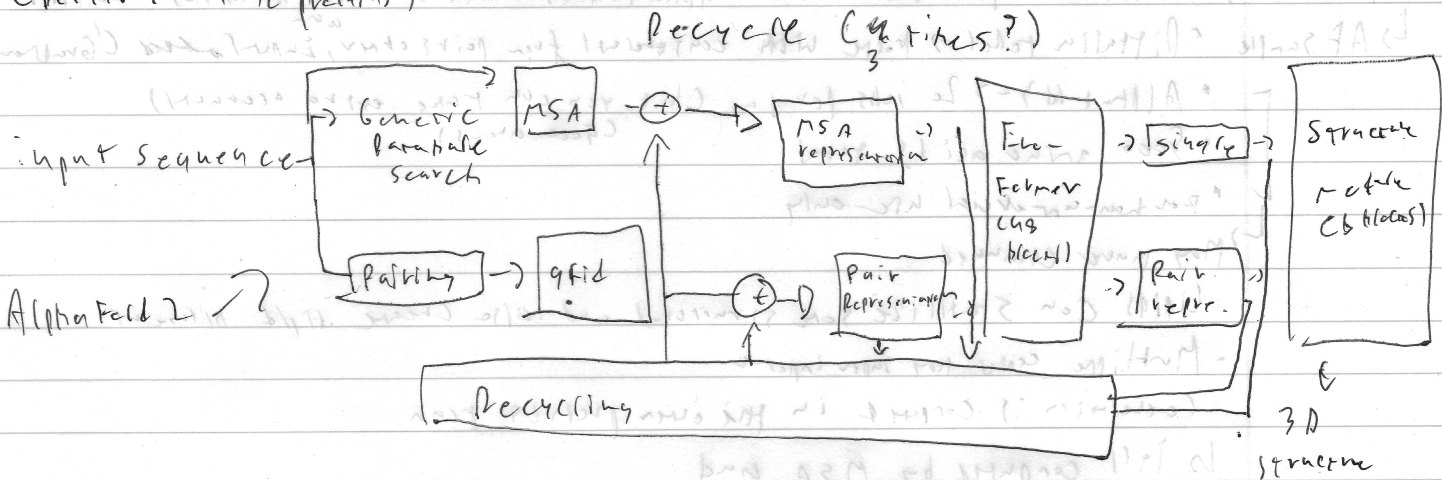
AlphaFold 3 -> Sergey - talk 10/9/24

Background

- ↳ Utilize evolutionary information to search against a database of sequences
- ↳ The why behind MSA.



(better in large proteins)



Evolutionary Triangular Inequality

- ↳ $z < x + y$ (vertices & edges and corresponding edges in a graph)
- Overlap product mean -> All correlation happens
- Ablations led to mutual changes in AlphaFold2 but showed changes made large differences
- Additionally, for "fold" of proteins there were hierarchical graphs by function but by mean)
- ↳ Triangular attention can learn to enforce Triangular Inequality

The structure module almost started at 0,0,0 (and it approximates MPS) (69%)

- ↳ Limitations: potential state (same constraints, same solution)
- ↳ Structure module takes a short time to run but you could only change MSA

AlphaFold2 couldn't deal with extremely large proteins in 8 steps

- ↳ AlphaFold3, recycles earlier, Diffusion module
- Evolver -> path forward (in AlphaFold3)

Diffusion: Adding noise to given a model to undo the noise

- Instead of 0,0,0 you start at noise and then use 2 steps to denoise (diffusion sampled)
- Completely random noise by parameters (typically doesn't matter)